

harming her. On the other hand, if it insisted that she take the medicine, the robot would be impinging on the person's autonomy.

To help Nao around such problems, scientists and philosophers provided it with examples – case studies in which such conflicts have been resolved by bioethicists. From these algorithms, patterns were found which could help guide the robot through such situations.

However, there's a catch: the principles that are derived from the algorithm are not written in a computer code. This means that you wouldn't know the reason why a program may tell the robot something is ethically right or not.

Many engineers think that, in such scenarios, it's best to create programs that have the rules clearly formulated, as opposed to asking the robot to derive its own rules. The proponents of the rule-based approach cites one advantage above all else: you are always clear as to why the robot makes a particular choice. It's particularly important in fields like the military, where having a robot drawing its own ethical conclusions on the field could be disastrous.

The machine-learning approach, on the other hand, brings forth the possibility of robots which could learn from experience. This could help create more flexible robots and could be more useful than the strictly programmed variants. Many roboticists believe that the ideal way forward would involve combining both the approaches. But combining the two approaches in a workable way is a challenge that is yet to be met.

Notwithstanding the manner in which the issue would be tackled, it looks more and more as though a workable solution wouldn't be confined to the three laws that Asimov put forward. The reality, as it's turning out, could be far more complex than anyone ever imagined.

The BS8611 Robots and robotic devices

The British Standards Institute(BSI) issued a document called BS8611 Robots and robotic devices. Drafted in a language that's a bit too dry, it's a health and safety manual for manufacturers of robots. Notwithstanding its colourless language, the manual covers a whole lot of topics that could have been taken from science fiction. Topics like robot deception, robot addiction, and even the possibility of self-learning mechanisms exceeding their limits, falls under 'hazards which should be considered by the manufacturers.'

It's probably the first ever published guideline for the ethical design of robots. Some consider it the first step in inculcating ethics in robots.